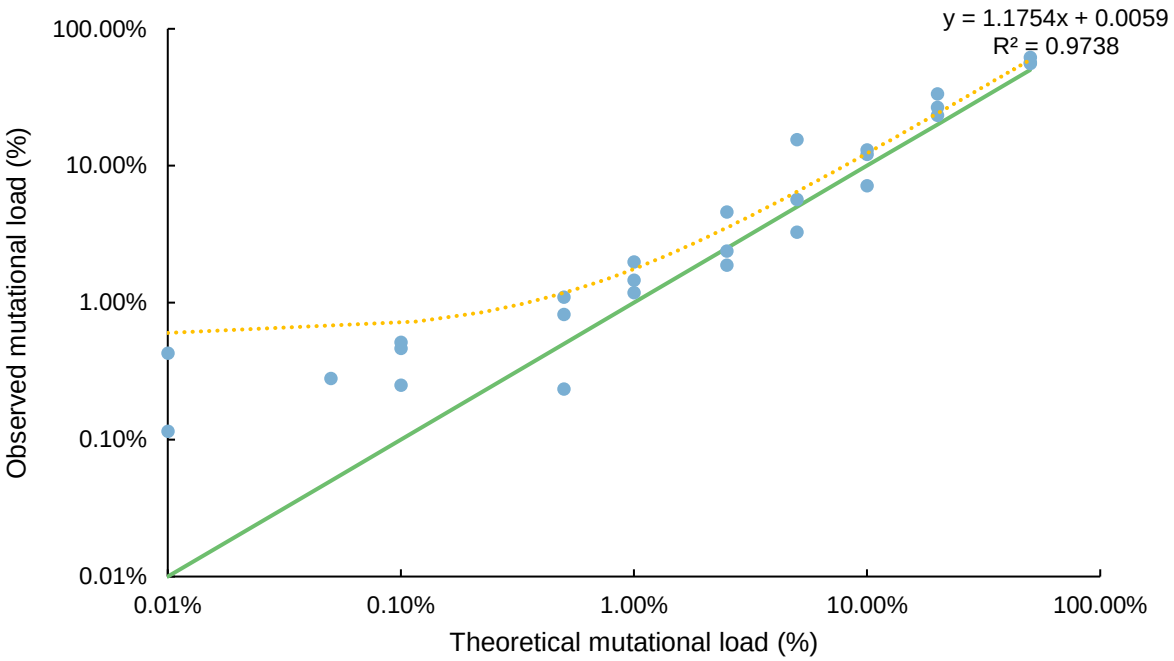
Supplemental File 3

Supplemental Figure S1:



Expected versus observed mutational load of titration series of synthetic mixture samples using the Naica system. The green line represents the expected mutational load, the blue dots represent the observed mutational loads, resulting from single measurements. The orange dotted line represents the linear trendline.

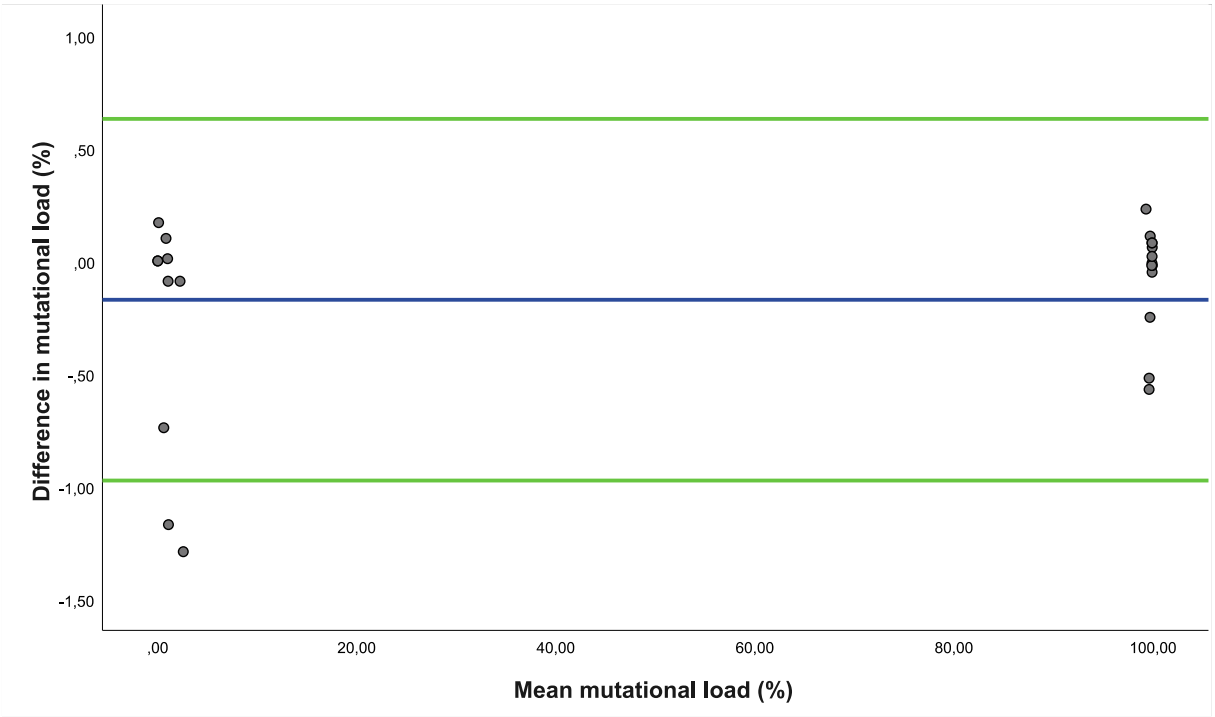
Supplemental Table S2: Mutational load as obtained by Naica ddPCR for all samples, along with the 95% confidence interval, the number of analyzed droplets, the number of positive droplets per allele, the mutational load as determined by NGS, and the difference between both measurement techniques.

Sample	# Analyzed droplets	# Positive wild-type droplets	# Positive mutant droplets	Mutational load ddPCR (%)	95% CI upper bound	95% CI lower bound	Mutational load (NGS) (%)	Difference in mutational load (%)
1	15546	2	1432	99.87	99.97	99.31	99.09	0.78
2	23322	14	1595	99.16	99.48	98.64	99.13	0.03
3	22592	1	1050	99.91	99.91	99.64	99.16	0.75
4	24219	3	3527	99.92	99.96	99.85	99.05	0.87
5	22372	2	1554	99.88	99.95	99.67	99.14	0.74
6	23438	4	1914	99.80	99.92	99.48	99.06	0.74
7	25024	3	2242	99.87	99.94	99.73	99.03	0.84
8	21984	3	1864	99.85	99.94	99.59	99.02	0.83
9	23247	2	1395	99.86	99.95	99.64	99.1	0.76
10	22049	5	1358	99.64	99.84	99.21	98.89	0.75
11	26688	8	4486	99.84	99.91	99.70	99.01	0.83
12	24758	13	7008	99.84	99.91	99.73	98.9	0.94
13	25292	5	3534	99.87	99.95	99.67	99.03	0.84
14	15070	592	0	0.00	0.00	0.00	0.3	-0.30
15	21747	1174	0	0.00	0.00	0.00	0.28	-0.28
16	23800	2043	0	0.00	0.00	0.00	0.25	-0.25
17	15289	255	6	2.28	4.98	1.03	2.21	0.07
18	23327	984	17	1.66	2.60	1.06	1.92	-0.26
19	21171	1926	22	1.08	1.62	0.71	1.16	-0.08
20	20027	2284	24	0.98	1.44	0.67	1.54	-0.56
21	21081	2846	24	0.78	1.16	0.53	1.18	-0.40
22	20043	739	25	3.21	4.65	2.21	2.97	0.24
23	23864	3312	35	0.97	1.34	0.70	0.96	0.01

Supplemental Table S3: Technical replicates of 6 samples, along with the average mutational load and the calculated between-run standard deviation.

Sample code	Mutational load 1 (%)	Mutational load 2 (%)	Mutational load 3 (%)	Average mutational load (%)	Standard deviation (%)	Coefficient of variation (%)
1	99.87	99.49	99.07	99.48	0.40	0.40%
2	99.16	99.07	99.03	99.09	0.07	0.07%
14	0.00	0.00	0.00	0.00	N/A	N/A
18	1.66	1.89	1.42	1.66	0.23	14.16%
22	3.21	4.03	3.36	3.53	0.44	12.36%
23	0.97	0.62	0.76	0.78	0.18	22.80%

Supplemental Figure S2



Bland-Altman comparing Naica and Bio-Rad ddPCR for mtDNA heteroplasmy assessment. The x-axis shows the mean mutational load, the y-axis shows the difference between both measurement techniques. The blue reference line represents the average difference between both measurement techniques, the green reference lines represent the 95% CI around this difference.

Supplemental Table S4: Number of positive droplets observed for all NTCs.

NTC Number	Platform	# WT positive droplets	# Mutant positive droplets
1	Bio-Rad	4	0
2	Bio-Rad	3	3
3	Bio-Rad	1	0
4	Bio-Rad	0	0
5	Bio-Rad	1	0
6	Bio-Rad	2	0
7	Bio-Rad	6	1
8	Bio-Rad	2	0
9	Bio-Rad	1	0
10	Bio-Rad	4	0
11	Bio-Rad	4	0
12	Naica	1	1
13	Naica	3	2
14	Naica	2	1
15	Naica	2	0
16	Naica	1	0

Supplemental Figure S3: Two example plots of NTCs (left: Naica, right: Bio-Rad).

